

The Tinkerer's Manual

How to program Gaussian splats: what one splat is, how the toys are built, and a small language for writing your own splat equations, and toys that show pictures and read what you type.



Every toy in [Splashery](#) is made of Gaussian splats: tens of thousands of soft, colored blobs that blend into a solid-looking thing. This manual is for anyone who knows some math and wants to make splats do what they say. It goes level by level. You can stop after any level and still have something to play with.

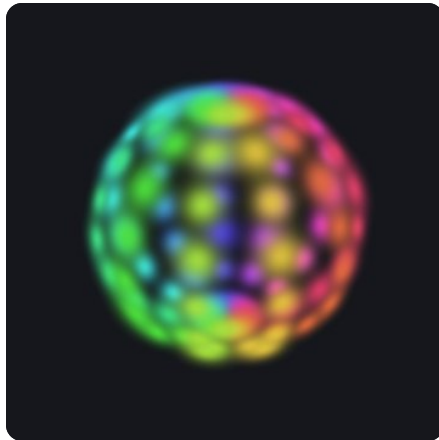
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LEVEL 1

What a splat is

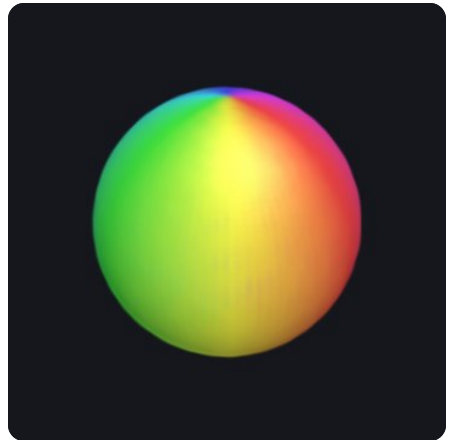
A **splat** is a small cloud of color floating in space. Up close it looks like a soft ellipse, bright in the middle and fading at the edges. On its own it is nothing much. Put thousands side by side and they melt into surfaces: a torus, a seashell, a cat.



100 splats



400 splats



6,000 splats

The same sphere with more and more splats, made with the Splat equation toy. With few splats you see each soft blob; with enough of them, they blend into one smooth surface.

The numbers in one splat

Each splat is described by a handful of numbers:

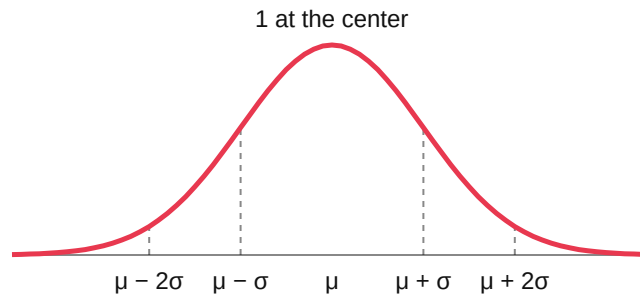
Part	Numbers	What it means
Center μ	3: x, y, z	Where the middle of the splat is.
Sizes s	3: s_1, s_2, s_3	How far it spreads along each of its own three axes. Equal sizes make a ball; one small size makes a flat disc.
Rotation q	4 (a quaternion)	Which way those three axes point.
Color	3: red, green, blue	Each from 0 to 1.
Opacity α	1	How solid its middle is, from 0 (clear) to 1 (solid).

That is 14 numbers per splat. A Splashery toy has about 60,000 splats on a small phone, up to 280,000 on a strong computer, and up to 400,000 on the strongest devices (the site picks for you). Many splat files also let a splat's color change a little with the direction you look from; the toys you build here use one color per splat.

The Gaussian: a bell in three dimensions

The "Gaussian" in Gaussian splat is the bell curve, named after the mathematician Carl Friedrich Gauss. In one dimension, a bell centered at μ with width σ (sigma, the standard deviation) is

$$G(x) = \exp(-(x - \mu)^2 / 2\sigma^2)$$



The bell curve. At one σ from the center it has fallen to about 0.61, at two σ to about 0.14, and at three σ to about 0.01, so a splat fades out within about three of its sizes.

In three dimensions the width becomes a 3-by-3 matrix Σ (capital sigma), called the **covariance**, and the squared distance becomes a quadratic form. For a point p in space, the splat's strength there is

$$G(p) = \exp(-1/2 (p - \mu)^T \Sigma^{-1} (p - \mu))$$

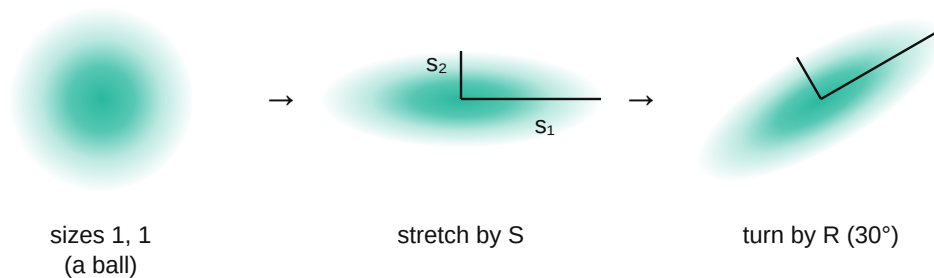
Where G is the same value, you are on the same "shell" around the center, and those shells are ellipsoids: stretched and turned spheres.

The covariance, built from sizes and a rotation

Not every 3-by-3 matrix is a valid covariance. It has to be symmetric, and it has to be positive definite (the bell must fall off in every direction). Instead of picking its nine numbers directly, a splat builds it from its three sizes and its rotation:

$$\Sigma = R S S^T R^T, \text{ where } S = \begin{pmatrix} s_1 & 0 & 0 \\ 0 & s_2 & 0 \\ 0 & 0 & s_3 \end{pmatrix}$$

and R is the rotation matrix of the quaternion q . Read it right to left, as a recipe for a shape: start with a round ball of size 1, stretch it by s_1 , s_2 and s_3 along x , y and z , then turn it by R . Any sizes above zero and any rotation give a valid covariance, which is exactly what you want when a computer is nudging millions of numbers.



A covariance from sizes and a rotation, drawn in two dimensions: a round splat is stretched along its own axes, then turned.

A worked example. Take sizes 0.3, 0.1 and 0.1, turned 30° about the z axis. Then SS^T has 0.09, 0.01 and 0.01 on its diagonal, and multiplying out $R SS^T R^T$ gives

$$\Sigma \approx \begin{pmatrix} 0.070 & 0.035 & 0 \\ 0.035 & 0.030 & 0 \\ 0 & 0 & 0.010 \end{pmatrix}$$

The top-left number is $0.09 \cos^2 30^\circ + 0.01 \sin^2 30^\circ = 0.07$, and the ones beside it are $(0.09 - 0.01) \sin 30^\circ \cos 30^\circ \approx 0.035$. Those off-diagonal numbers are the tilt.

From 3D to the screen

To draw a splat, the renderer works out what its bell looks like from the camera. It first turns the covariance into the camera's view, with the camera's rotation W . The perspective of a camera is not a straight-line map, but close to one point it nearly is, and its best straight-line stand-in there is a matrix J (the Jacobian of the projection). Then the splat's covariance on the screen is

$$\Sigma' = J W \Sigma W^T J^T$$

keeping only its top-left 2-by-2 block. That 2-by-2 matrix is a flat bell on the screen: an ellipse. Its eigenvectors are the directions of the ellipse's axes, and the square roots of its eigenvalues are how far it spreads along them. This trick comes from EWA splatting (Zwicker, Pfister, van Baar and Gross, 2001).

Sorting and blending, back to front

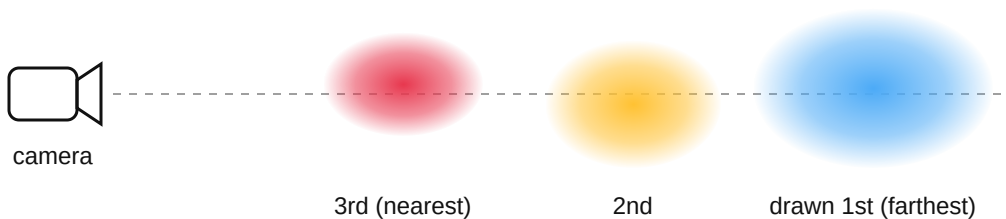
Splats are see-through at their edges, so the order they are drawn in matters. Each frame, the splats are sorted by how far they are from the camera, and painted from the farthest to the nearest. At each pixel, a splat's strength there is its opacity times its bell, $a = \alpha G'$, and it is laid over what is already there:

$$C \leftarrow a c + (1 - a) C$$

where c is the splat's color and C the pixel's color so far. Written out for all the splats at once, counting from the nearest ($i = 1$) to the farthest, the pixel's color is

$$C = \sum_i c_i a_i \prod_{j < i} (1 - a_j)$$

Each splat shows as much as the splats in front of it let through. This is how the 2023 paper that started the craze, "3D Gaussian Splatting for Real-Time Radiance Field Rendering" (Kerbl, Kopanas, Leimkühler and Drettakis), draws its scenes, and it is how Splashery draws every toy.



Splats are painted from the back to the front, each one laid over the ones behind it.

Why the order matters in the toys. Splashery sorts the splats in the pose a toy was built in, so a piece that moves keeps the order it had at rest. Turn a solid thing more than a quarter turn and its far side can be drawn over its near side. The toys work around it: the Splat equation toy, for one, builds its shape at twelve moments and shows the copy nearest the moment you are at (Level 3).

LEVEL 2

How a toy is built

Nobody places Splashery's splats by hand. Most toys are **recipes**: short programs that describe a toy with a kit of shapes, and the kit fills them with splats when you open the toy. A recipe is a few kilobytes of code instead of megabytes of splats, and because every random choice comes from a seed, it builds the same toy on every device.

Recipes are written in JavaScript. This level is a tour, so you can read one; the complete guide is [docs/PACKS.md](#) in the source code.

Two places a toy lives

A toy's name and shelf go in a **catalog** entry (`src/toys.js`), and the toy itself in a **recipe** in a pack file (`src/packs/...js`):

```
{
  id: "little-windmill",      // lower-case words joined by hyphens
  label: "Little windmill",   // its name on the shelf
  category: "objects",        // which shelf
  kind: "kit",
  pack: "my-pack",            // loads src/packs/my-pack.js
}
```

A recipe has six basic parts, all optional except `build` (Level 5 adds `pictures` , `input` and `tiltLock` , and [PACKS.md](#) has more, such as `grab` and `drag` for touch play):

<code>build(k, o)</code>	Describes the toy with the kit <code>k</code> . <code>o</code> holds the chosen options.
<code>options</code>	Choices that rebuild the toy: a color, a style, a number of petals. They are saved in links.
<code>controls</code>	Live values from 0 to 1: sliders, toggles that ease on and off, and pulses that jump to 1 and fall back.
<code>action</code>	Which control a tap on the toy fires.
<code>drive(t, c, out)</code>	Runs every frame: from the time <code>t</code> and the controls <code>c</code> , it sets how the parts move.
<code>alive</code>	<code>true</code> when the toy moves by itself, so frames keep coming.

Shapes from the kit

`k.sphere(r)`, `k.ellipsoid(a, b, c)`, `k.box(w, h, d)`,
`k.roundedBox(w, h, d)`, `k.cylinder(r, h)`, `k.cone(r0, r1, h)`,
`k.torus(R, r)`, `k.disc(r)`, `k.lathe(profile)` (a vase or a bottle, turned around an axis), `k.tube(curve, radius)` (a stem, an arm, a knot) and
`k.param((u, v) => [x, y, z])`, any surface you can write in two variables (the same idea as Level 3; here *u* and *v* each run from 0 to 1), and
`k.radial((dir) => r)`, a blobby surface given by its distance from the middle in every direction. Nothing is drawn until you add a shape:

```
k.add(k.sphere(0.2), { pos: [0, 1, 0], color: "#ffcc33", weight: 2 });
```

The kit shares the splats out between the shapes by their surface area, so the toy is evenly dense; `weight` gives a small shape more (or `share` a fixed fraction of all the splats). For free-form splats, like stars, sparks and crumbs, `k.cloud` places them one by one. After `build`, the kit centers the toy and scales it to fit the view, so you can model in any units.

Colors from functions

A color can be a fixed one (`"#9c3b2c"` , or `[r, g, b]` with each from 0 to 1; there are no color names) or a function that the kit asks for every splat. It gets the splat's place `c.p`, its surface direction `c.n`, its surface coordinates `c.u` and `c.v`, and a random number and noise that come from the seed. Return `null` to leave a hole:

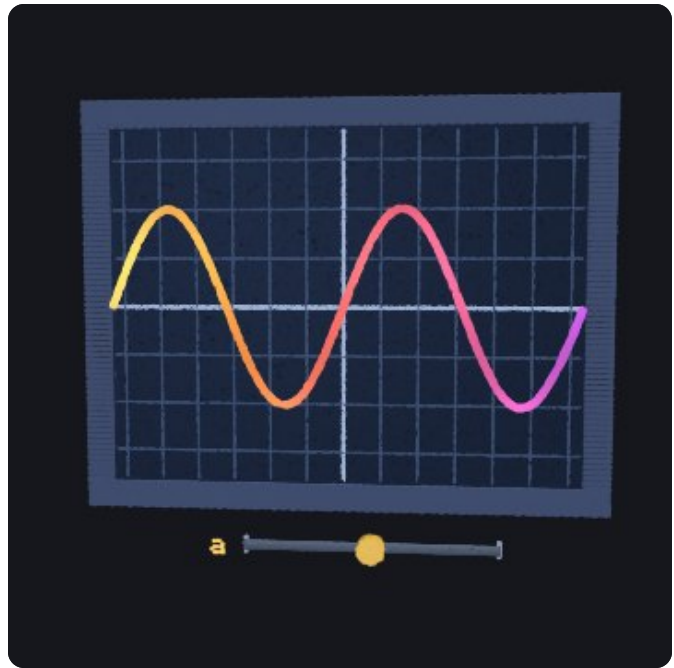
```
color: (c) => (c.p[1] > 0.5 ? "#ffffff" : "#3a7d44"), // snow above, grass below
```

Parts that hinge, spin or slide

A **part** is a group of shapes that moves as one solid piece: `k.part("lid", { pivot, axis })`. Every frame, `drive` tells each part where to be: an `angle` about its axis (a lid on a hinge, a wheel that spins), an `offset` (a drawer that slides), a `scale`, or `visible`. A toy may have up to 15 parts. The treasure chest's lid is a part on a hinge.



The treasure chest: its lid is a part on a hinge.



The graph plotter: the a slider is a control.

Behaviors

Some motion is too fine for parts: a flame that flickers, snow that falls, grass that sways. A **behavior** moves each splat on its own on the graphics card: `kind: "flame", "twinkle", "sway", "orbit", "fall", "wave"` and more. **Morphs** give each splat a second place, `to: (c) => [x, y, z]`, and slide it there as one of four channels goes from 0 to 1. The Splat equation toy's motion is made of morphs.

A small, complete recipe

Here is a whole toy: a little windmill whose sails turn with a Wind slider, and a tap that blows a gust, spinning them once more around.

```
import { mix } from "../src/kit.js";

// A little windmill: a stone tower, a red cap, and four sails that turn in
// the wind. A tap blows a gust that spins them once more around.
export const RECIPES = {
  "little-windmill": {
    alive: true,
    options: [{ key: "sails", label: "Sail color", type: "color", default: "#f2eee4" }],
    controls: [
      { key: "wind", label: "Wind", type: "slider", default: 0.4 },
      { key: "gust", label: "Gust", type: "pulse", ease: 3 },
    ],
    action: { key: "gust", label: "Blow a gust" },
  },
}
```

```

drive(t, c, out) {
  // The wind turns the sails; a gust adds one extra turn, eased in and
  // out, so they end where they would have been anyway.
  const g = c.gust > 0 ? 1 - c.gust : 0;
  const extra = 2 * Math.PI * g * g * (3 - 2 * g);
  out.parts.sails = { angle: t * (0.3 + 2 * c.wind) + extra };
},
build(k, o) {
  // The tower: whitewashed stone, darker toward the ground.
  k.add(k.cone(0.5, 0.3, 1.6), {
    color: (c) => mix("#8b8f97", "#ece6d8", (c.p[1] + 0.8) / 1.6),
    interior: 0.1,
    core: "#6b5b4b",
  });
  // The cap.
  k.add(k.cone(0.36, 0.02, 0.4), { pos: [0, 1.0, 0], color: "#9c3b2c" });
  // The sails turn together, about an axis that points at you.
  const sails = k.part("sails", { pivot: [0, 0.6, 0.42], axis: [0, 0, 1] });
  k.add(k.sphere(0.07), { pos: [0, 0.6, 0.42], part: sails, color: "#4b3a2a" });
  for (let i = 0; i < 4; i++) {
    const a = (i * Math.PI) / 2;
    k.add(k.box(0.16, 0.9, 0.02), {
      pos: [-0.5 * Math.sin(a), 0.6 + 0.5 * Math.cos(a), 0.44],
      rot: [0, 0, (a * 180) / Math.PI],
      part: sails,
      color: o.sails,
    });
  }
},
},
},
};

```



This recipe, built by the kit.



The Windmill on the shelf, a bigger toy built the same way.

Read it from the bottom up. `build` adds a cone for the tower (its color a function of height), a smaller cone for the cap, and a part called "sails" with a hub and four boxes, each turned a quarter turn more than the last. `drive` turns the sails part: steadily with the wind, plus one extra turn while the gust pulse runs, eased with the smoothstep curve $3g^2 - 2g^3$ so it starts and ends gently. A full extra turn lands the sails where they would have been anyway, so nothing jumps when the gust ends.

LEVEL 3

The splat equation language

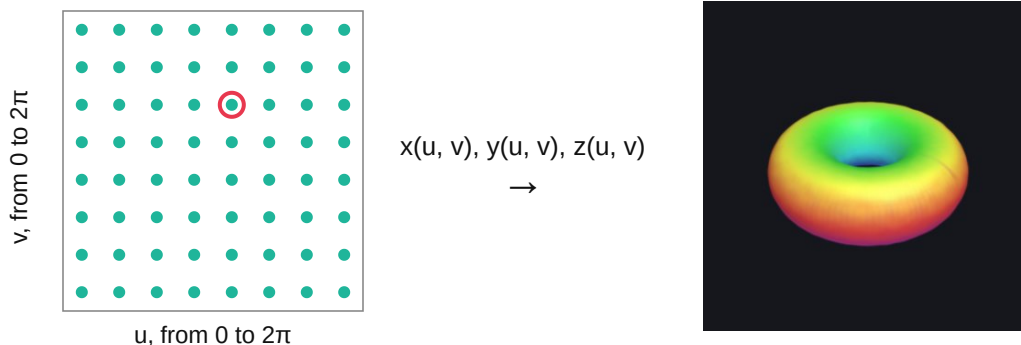
Recipes are powerful, but they are code, and code from a stranger can do things you did not ask for. The **Splat equation** toy on the math shelf gives you a smaller language that is only math. You write where every splat goes and what color it is, as equations, and nothing else can happen. That is why a link with your equations in it is safe to share.

Open it with any link in the gallery (Level 4). Type in the box under **Your own splat program** in the Toy tab and press **Splat it**. The Program list has eight famous shapes to start from.

The idea: every splat gets its own u and v

The toy lays out its splats on a grid of two numbers, u and v , each running over a range you choose. Your equations turn each splat's (u, v) into a place in space, (x, y, z) . If u goes around a circle and v around a smaller circle, the grid wraps into a torus:

```
x = (2 + cos v) * cos u
y = sin v
z = (2 + cos v) * sin u
u = 0 .. 2pi
v = 0 .. 2pi
```



Each dot on the grid is one splat with its own (u, v) . The equations carry every dot to its place on the torus.

The fields

A program is a list of **fields**, each written `name = value`. You can type one field or several, split by `;` (a text file can put one on each line). Any field you leave out

keeps what is showing.

Field	What it is	If left out
<code>x</code> , <code>y</code> , <code>z</code>	Where each splat goes, as equations in u , v and t . y points up.	Needed
<code>u</code>	The range of u , like <code>0 .. 2pi</code> .	Needed
<code>v</code>	The range of v .	<code>0 .. 1</code>
<code>hue</code>	Color as a place on the color wheel: 0 is red, 1/6 yellow, 1/3 green, 1/2 cyan, 2/3 blue, 5/6 magenta, and 1 is red again. Any number works; only the part after the point counts.	Use r , g , b
<code>r</code> , <code>g</code> , <code>b</code>	Color as red, green and blue, each from 0 to 1. Setting any of them turns <i>hue</i> off; setting <i>hue</i> turns them off.	0.5 each (gray)
<code>size</code>	How big each splat is, in the same units as x , y and z : about the σ of its bell. It may use u , v and t .	<code>0.05</code>
<code>count</code>	How many splats: 100 to 10,000.	<code>4000</code>
<code>spread</code>	<code>grid</code> spaces the splats evenly; <code>random</code> scatters them at random over the ranges (a galaxy, a firework).	<code>grid</code>

You can also write `color` for `hue` , `red` , `green` and `blue` for `r` , `g` and `b` , `splats` for `count` , `x(u,v,t)` for `x` , and add notes after a `#` . Names can be in capitals or not. A range can be written `0 .. 2pi` , `0 to 2pi` or `0, 2pi` .

Expressions

The equations are read by the same small, safe reader as the Graph and Surface plotters. It understands:

- **Numbers:** `2` , `0.5` , `.25` .
- **Variables:** u , v and t . Range ends and the count are numbers only.
- **Constants:** `pi` (or π) and `e` .
- **Operators:** `+` `-` `*` `/` `^` , written `+` `-` `*` `/` `^` (and `**` , `.` , `^2` , `^3`), with brackets `( )` and `|x|` for absolute value.
- **Functions:** `sin` `cos` `tan` `asin` `acos` `atan` `sinh` `cosh` `tanh` `exp` `log` (natural; `ln` too) `log10` `sqrt` (or $\sqrt{}$) `abs` `floor` , and `min` and `max` of two or more values.

- **Shortcuts:** `2u` means $2 \times u$, `3sin(u)` and `(u + 1)(v - 1)` multiply, and a function can skip its brackets on a simple product: `sin 2u`, `cos v`.

Powers go first, then $\times$ and $\div$, then $+$ and $-$, and a power binds to the right: `2^3^2` is 2^9 , and `-u^2` is $-(u^2)$. Each field can be up to 120 characters long.

The grammar

For the curious, here is the whole grammar. Words in quotes are typed as they are.

```

program    = field "=" value { ( ";" | new line ) field "=" value }
field      = "x" | "y" | "z" | "u" | "v" | "hue" | "r" | "g" | "b"
           | "size" | "count" | "spread"
value      = sum (x, y, z, hue, r, g, b, size)
           | sum ".." sum (u, v: numbers only)
           | sum (count: a number only)
           | "grid" | "random" (spread)
sum        = product { ( "+" | "-" ) product }
product    = signed { ( "*" | "/" ) signed | power } (nothing between: 2u)
signed     = { "-" | "+" } power
power      = atom [ "^" signed ]
atom       = number | "u" | "v" | "t" | "pi" | "e"
           | function "(" sum { "," sum } ")"
           | function atom { atom } (sin 2u)
           | "(" sum ")" | "|" sum "|"

```

Time

t is time. A tap on the toy plays it from 0 to 2π in about 4 seconds, and the **t** slider in the Toy tab holds it anywhere in between. Anything with t in it moves: `u + t` turns a shape, `sin(t)` breathes, and `sin(3u - t)` sends a wave around.

Behind the scenes, the toy builds your shape at 12 moments, $t = 0, \pi/6, 2\pi/6$ and so on, and each copy slides every splat in a straight line to its place in the next copy (a morph, Level 2), so the motion is exact at those 12 moments and smooth in between. A shape that turns dips very slightly inward between moments, about 3 percent.

Make it loop. Because t ends at 2π , a shape made of `sin(t)`, `cos(t)` and angles like `u + t` ends exactly where it began. A shape with plain `t` in it (like `y = t`) jumps back at the end. `sin(t/2)` goes out and back once; `sin(2t)` twice.

A program without t stands still. Then a tap clears it and draws the splats again one by one, in the order they are made: the first value of u with every v , then the next u , and so on.

What the toy does for you

- **It fits the view.** The whole shape (at every moment of t) is scaled to fit, so your units don't matter; only the sizes compared with the shape do.
- **It spaces the grid.** With `spread = grid`, the side of the shape that is longer gets more splats, so they sit about evenly apart. A shape that uses only u is a curve, and all the splats go along it.
- **It lights surfaces.** For a surface, each splat is a flat disc facing out from it (along the cross product of its slopes in u and v), lit from above. The **Light and shade** switch turns the lighting off, so you see your colors exactly.
- **It skips impossible points.** Where an equation has no value (the square root of a negative number, a division by zero), that splat is left out. So are points that shoot off far beyond the rest.
- **It explains mistakes.** Each field that can't be read gets one plain message, like “z: *That can't be drawn: I don't know “q”.*”, and the shape on the stage stays as it was.

Why it is safe

The reader never runs what you type. It breaks the text into numbers, the letters u , v and t , the names in the lists above and the signs, and builds a small tree of arithmetic from them. There is no way to name anything else, so the worst a program can do is draw a strange shape. Your program lives in the toy's link ("Copy link" in the Share tab), so you can send it to anyone.

LEVEL 4

A gallery of programs

The first eight are the toy's own programs; the rest are small ideas to copy and change. Each button opens the toy with that program; then tap it to play t . While the toy is being tried out, it is on the shelf only when Splashery is opened with `? labs=1`, but these links always open it.

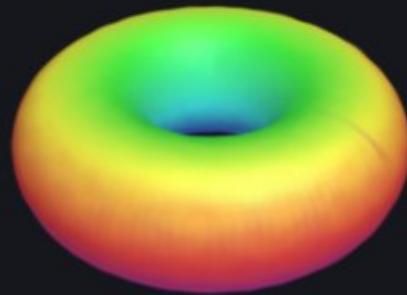


Sphere

A sphere from two angles, u around and v from pole to pole, spinning with $u + t$. Six stripes from $\text{floor}(6u/(2\pi))$.

```
x = sin(v)cos(u + t)
y = cos(v)
z = sin(v)sin(u + t)
u = 0 .. 2pi
v = 0 .. pi
hue = floor(6u/(2pi))/6 + 0.04
size = 0.035
count = 7000
```

Open it in the toy



Torus

A tube around a ring. $v + t$ rolls the tube around itself, so it turns inside out as you watch.

```
x = (2 + cos(v + t))cos(u)
y = sin(v + t)
z = (2 + cos(v + t))sin(u)
u = 0 .. 2pi
v = 0 .. 2pi
hue = v/(2pi)
size = 0.09
count = 6000
```

Open it in the toy

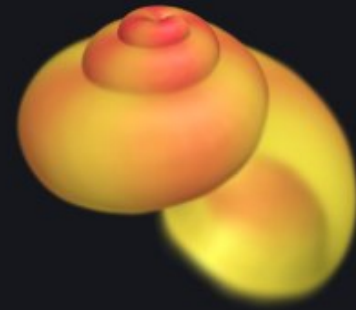


Möbius strip

A band with a half twist: $\cos(u/2)$ turns the band over once as u goes around. The colors flow along it.

```
x = (1 + v cos((u + t)/2))cos(u + t)
y = v sin((u + t)/2)
z = (1 + v cos((u + t)/2))sin(u + t)
u = 0 .. 2pi
v = -0.4 .. 0.4
hue = u/(2pi)
size = 0.035
count = 4000
```

[Open it in the toy](#)



Seashell

A tube that grows by $\exp(0.1u)$ as it winds three times around, turning with $u + t$. Its splats grow too.

```
x = exp(0.1u)cos(u + t)(1.2 + cos v)
y = exp(0.1u)(sin(v) - 2.4) + 3
z = exp(0.1u)sin(u + t)(1.2 + cos v)
u = 0 .. 6pi
v = 0 .. 2pi
hue = 0.02 + 0.1u/(6pi) +
0.03cos(4u)
size = 0.14exp(0.1u)
count = 10000
```

[Open it in the toy](#)

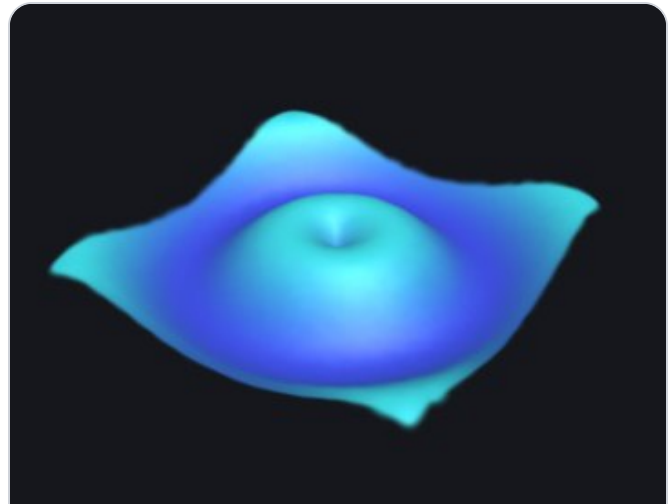


Trefoil knot

The simplest real knot, a curve in u alone, turned about the vertical by the rotation formulas in t .

```
x = (sin(u) + 2sin(2u))cos(t) -
sin(3u)sin(t)
y = cos(u) - 2cos(2u)
z = -(sin(u) + 2sin(2u))sin(t) -
sin(3u)cos(t)
u = 0 .. 2pi
v = 0 .. 1
hue = u/(2pi)
size = 0.16
count = 2500
```

Open it in the toy



Wave

A flat sheet in u and v with ripples running out from the middle: $\sin(\text{distance} - t)$.

```
x = u
y = 0.45sin(2sqrt(u^2 + v^2) - t)
z = v
u = -3 .. 3
v = -3 .. 3
hue = 0.58 - 0.07sin(2sqrt(u^2 +
v^2) - t)
size = 0.07
count = 6000
```

Open it in the toy

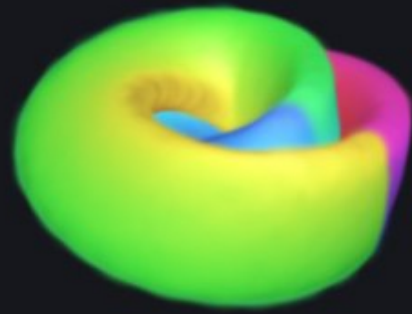


Spiral galaxy

Two arms from $\text{floor}(v)$ (0 or 1, half a turn apart), scattered at random, turning once. Colors from r, g and b.

```
x = u^1.5 cos(pi floor(v) + 5u +
1.2(v - floor(v)) - t)
y = 0.06(1 - u)sin(91v + 37u)
z = u^1.5 sin(pi floor(v) + 5u +
1.2(v - floor(v)) - t)
u = 0.02 .. 1
v = 0 .. 2
r = 0.55 + 0.45(1 - u)^2
g = 0.62 + 0.33(1 - u)^2
b = 1 - 0.45(1 - u)^3
size = 0.006 + 0.012(1 - u)^2
count = 10000
spread = random
```

Open it in the toy



Klein bottle (figure eight)

The figure-eight Klein bottle: a figure-eight tube around a ring, with a half twist, so it has no inside or outside.

```
x = (2 + cos(u/2)sin(v + t) -
sin(u/2)sin(2v + 2t))cos(u)
y = sin(u/2)sin(v + t) +
cos(u/2)sin(2v + 2t)
z = (2 + cos(u/2)sin(v + t) -
sin(u/2)sin(2v + 2t))sin(u)
u = 0 .. 2pi
v = 0 .. 2pi
hue = v/(2pi)
size = 0.13
count = 8000
```

Open it in the toy



Hello, circle

The smallest program: one loop of u makes a ring of 200 splats. No v , no t .

```
x = cos(u)
y = sin(u)
z = 0
u = 0 .. 2pi
hue = u/(2pi)
size = 0.06
count = 200
```

[Open it in the toy](#)

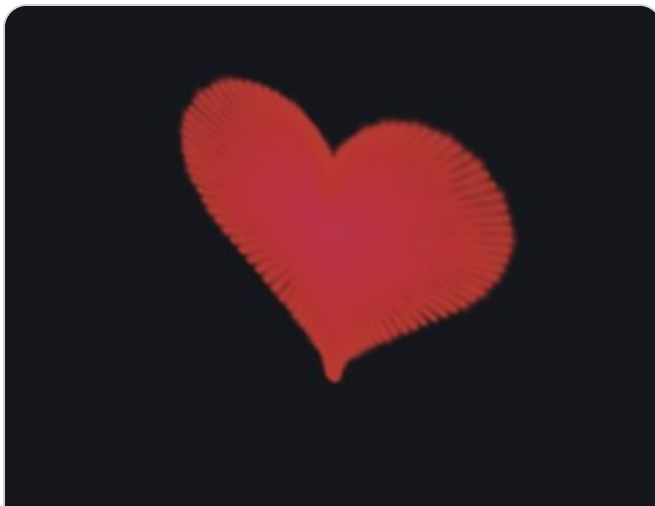


Bouncing spring

Three turns of a helix whose height breathes with $\sin(t)$.

```
x = cos(u)
y = (0.09 + 0.035sin(t))u - 0.9
z = sin(u)
u = 0 .. 6pi
hue = 0.08 + u/(6pi)/3
size = 0.07
count = 1500
```

[Open it in the toy](#)



Beating heart

A famous heart curve, filled in by v from the middle out, beating twice per cycle.

```
x = v(1 + 0.08sin(2t))16sin(u)^3
y = v(1 + 0.08sin(2t))(13cos(u) -
5cos(2u) - 2cos(3u) - cos(4u))
z = 0
u = 0 .. 2pi
v = 0 .. 1
hue = 0.97 + 0.04v
size = 0.35
count = 4000
```

[Open it in the toy](#)

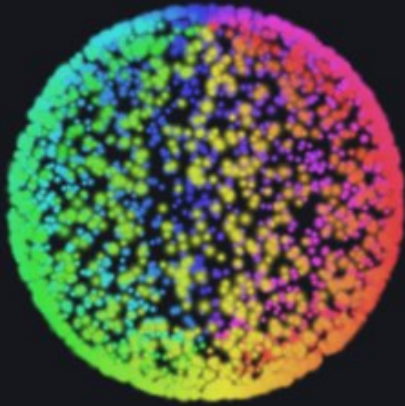


Rainbow ribbon

Color from r , g and b : three cosines a third of a turn apart make a rainbow along u .

```
x = u
y = 0.4sin(2u - t)
z = v
u = -3 .. 3
v = -0.4 .. 0.4
r = 0.5 + 0.5cos(u)
g = 0.5 + 0.5cos(u - 2)
b = 0.5 + 0.5cos(u + 2)
size = 0.05
count = 4000
```

[Open it in the toy](#)



Firework

Splats scattered at random over a sphere (v is the height, from -1 to 1) fly out, droop and come back.

```
x = (0.2 + sin(t/2))sqrt(1 -
v^2)cos(u)
y = (0.2 + sin(t/2))v -
0.35sin(t/2)^2
z = (0.2 + sin(t/2))sqrt(1 -
v^2)sin(u)
u = 0 .. 2pi
v = -1 .. 1
hue = u/(2pi)
size = 0.018
count = 3000
spread = random
```

[Open it in the toy](#)

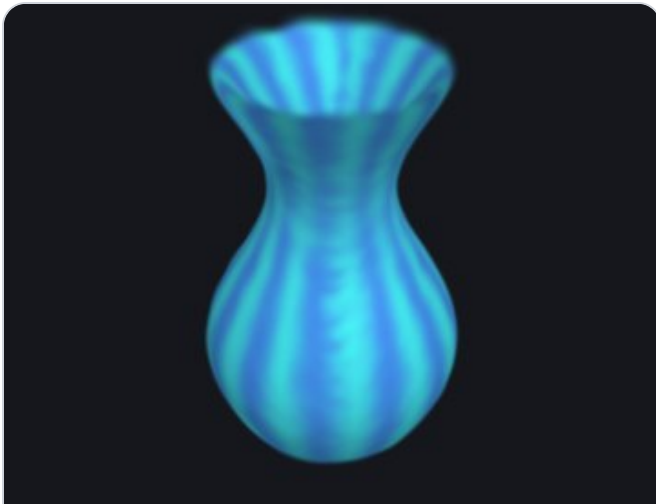


Opening flower

Five petals from $\cos(5u)$, and a cup that opens and closes with $\sin(t/2)$.

```
x = v(0.6 + 0.4cos(5u))cos(u)
y = (0.15 + 0.6sin(t/2))v^2
z = v(0.6 + 0.4cos(5u))sin(u)
u = 0 .. 2pi
v = 0 .. 1
hue = 0.92 - 0.12v
size = 0.035
count = 5000
```

[Open it in the toy](#)



Vase (no t)

A shape without `t` stands still, so a tap draws its splats again one by one, in the order they are made.

```
x = (0.5 + 0.2sin(3v))cos(u)
y = v
z = (0.5 + 0.2sin(3v))sin(u)
u = 0 .. 2pi
v = 0 .. 2.2
hue = 0.55 + 0.05sin(12u)
size = 0.045
count = 4000
```

[Open it in the toy](#)

Things to try. Change a number and see what moves. Swap `cos` and `sin`. Put `t` somewhere new. Make `count` small and `size` big to see the single splats. Try `spread = random` on a surface. Give a sphere a hue of `floor(8v/pi)/8`.

LEVEL 5

Pictures, screens and your own input

The toys in Level 2 are built from shapes. Some toys show something that is not a shape: a page of a PDF, a photo, a GIF, a video, or something you type or draw.

Three additions to a recipe make them: **picture sheets**, an **input panel** and a few **view settings**. They are still Gaussian splats, all the way down. The toys on the Pictures and pages shelf are being tried out, so they are on the shelf only when Splashery is opened with `?labs=1`.

Picture sheets: a page, a photo, a GIF, a video

A **sheet** is a flat place in a toy where a picture shows: `k.sheet({ id, center, width, height })`. The kit draws nothing there. After the build, the site opens the picture and turns it into splats on the sheet, in a background worker so the page stays smooth. A PDF page becomes a smooth sheet of large soft splats in the paper's color, with a small splat of its own color wherever the page differs from the paper, which is the ink (`method: "ink"`); in a photo, each pixel becomes a flat splat (`"pixels"`); a GIF or a video becomes a screen that is redrawn on every frame (`"screen"`). The site picks the method from the kind of file. It rebuilds a sheet finer as you zoom in, and frees the pages you have left, so a book of 200 pages does not need 200 pages of splats.



A PDF page (the Picture lab)



The recipe below: a photo in a frame

Both pictures are made of splats. Photo: "Tulip field" by DennisM2, CC0. The article is our own sample text.

Three parts of a recipe work together:

- `pictures`: { `sample`, `accept` } names what shows until you open your own (the sample's address) and which kinds a visitor may open: `"pdf"`, `"image"`, `"gif"`, `"video"`.
- `drive` may choose the page of each sheet with `out.sheets[id] = { page, visible }`. Pages count from 0, and a page that does not exist shows nothing. A sheet that is not visible is not built. Without it, every sheet shows the current page.
- `info.data.pictures` is the toy's view of the media: `page`, `count`, `kind`, `name`, `playing` and the functions `next()`, `prev()`, `go(n)` and `togglePlay()`.

A whole picture toy, in a few lines. Its `input` gives the Toy tab a button that opens a photo, a GIF or a video; the rest of the toy is a card and four wooden bars:

```
import { smoothstep } from "../src/kit.js";

// Two small picture toys for Level 5 of the Tinkerer's Manual: a framed photo
// that opens your own picture, and a little book whose page turns like paper.
export const RECIPES = {
  "little-frame": {
    turntable: false,
    tiltLock: true, // a drag only spins it left and right
    pictures: {
      sample: () => "assets/toys/picture-lab/photo.jpg", // shown until you open your own
      accept: ["image", "gif", "video"],
    },
  },
}
```

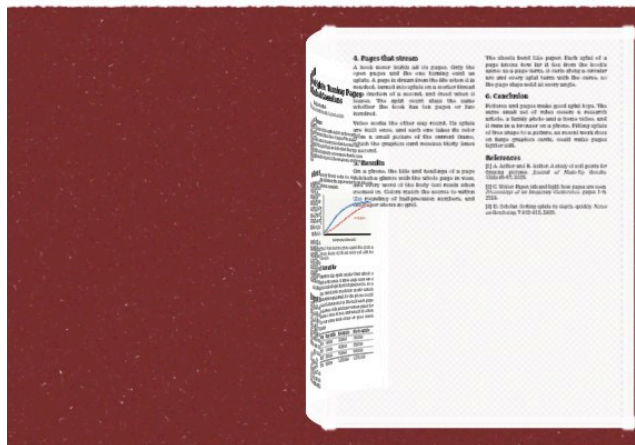
```

input: {
  title: "Your own picture",
  media: { accept: ["image", "gif", "video"] },
  note: "Open a photo, a GIF or a video. Files stay on this device.",
},
build(k) {
  // The picture: a sheet, filled with splats after the build.
  k.sheet({ id: "photo", center: [0, 0, 0.03], width: 1.5, height: 1 });
  // A gray card behind it, and four wooden bars around it.
  k.add(k.box(1.6, 1.1, 0.02), { pos: [0, 0, -0.02], color: "#9a9a9a" });
  for (const [w, h, x, y] of [
    [1.8, 0.1, 0, 0.6],
    [1.8, 0.1, 0, -0.6],
    [0.1, 1.1, -0.85, 0],
    [0.1, 1.1, 0.85, 0],
  ])
    k.add(k.box(w, h, 0.1), { pos: [x, y, 0], color: "#8a5a34" });
},
},

```

Turning a page

A page that turns like paper is a sheet with `leaf: slot` (a slot from 0 to 9) and a spine to turn about: `k.spine({ at, axis, dir })` gives the spine's place, its direction, and the direction from it along the pages when they lie flat. Each frame, `drive` sets `out.leaves[slot] = { angle, curl }`: the `angle` in radians (0 lying flat, π turned all the way over) and the `curl`, how much the page bends along a circular arc (a negative number lets the free edge lag behind, like paper in air). Every splat of the page turns with the curve at its place, so it stays a solid sheet at every angle, which a page bent with soft regions would not.



The little book below, halfway through a turn. The next page is already there under the turning one.

The little book below uses a pulse control for the turn, the way Level 2's gust does. The page under the turning one is built ahead of time (a sheet is built only while it is visible, so a page that shows up halfway would make the turn wait), and the page number moves on when the turn ends. Keep a turning sheet on the body (part 0), because a part's own turn would replace the leaf's. The far side of the turning page shows the page in mirror image, like tracing paper, until it lands.

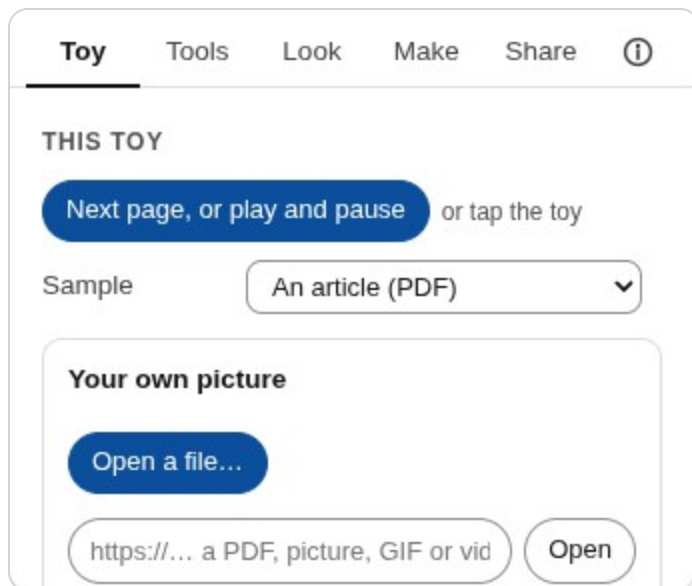
```
"little-book": {
  turntable: false,
  tiltLock: true,
  pictures: { sample: () => "assets/toys/picture-lab/article.pdf", accept: ["pdf"] },
  input: { title: "Your own book", media: { accept: ["pdf"] }, note: "Open a PDF." },
  controls: [{ key: "turn", label: "Turn the page", type: "pulse", ease: 1.2 }],
  action: { key: "turn", label: "Turn the page" },
  drive(t, c, out, info) {
    const pics = info.data?.pictures; // { page, count, go(n), next(), prev(), ... }
    // The page changes when the turn ends, and the turning page lands on the left.
    const turning = c.turn > 0;
    if (BOOK.turning && !turning && pics) pics.go(pics.page + 1);
    BOOK.turning = turning;
    const page = pics?.page ?? 0;
    // The pulse falls from 1 to 0: the page swings from 0 to a half turn (pi)
    // about the spine, and its free edge lags behind, like paper.
    const s = turning ? smoothstep(0, 1, 1 - c.turn) : 0;
    out.leaves = [{ angle: Math.PI * s, curl: -1.2 * Math.sin(Math.PI * s) }];
    out.sheets = {
      // Once the turn starts, the left page is the one that is about to land on it.
      left: { page: turning ? page : page - 1 }, // a page out of range shows nothing
      under: { page: page + 1 }, // built ahead, and uncovered as the page lifts
      front: { page }, // the turning page (its far side shows in mirror image)
    };
    // It is built, but hidden, until the turning page lands on it.
    out.parts.leftPage = { visible: turning ? 0 : 1 };
  },
  build(k) {
    BOOK.turning = false;
    k.spine({ at: [0, 0, 0], axis: [0, 1, 0], dir: [1, 0, 0] });
    const W = 1.4;
    const H = 1.9;
    const leftPage = k.part("leftPage");
    const page = { width: W, height: H };
    k.sheet({ id: "left", ...page, part: leftPage, center: [-W / 2, 0, 0], align: [1, 0] });
    k.sheet({ id: "under", ...page, center: [W / 2, 0, 0], align: [-1, 0] });
    // The turning page: a sheet on leaf slot 0, a hair in front of the others.
    k.sheet({ id: "front", ...page, center: [W / 2, 0, 0.008], align: [-1, 0], leaf: 0 });
    // The cover, behind the pages.
    k.add(k.box(2 * W + 0.1, H + 0.1, 0.03), { pos: [0, 0, -0.03], color: "#7a2f2f" });
  },
};

// Whether the page was turning in the last frame.
const BOOK = { turning: false };
```

Your own input: typing, drawing, opening a file

A recipe can ask for something from the visitor. Its `input` adds a panel to the Toy tab, in one of three forms:

- **Typing:** a text box and a button, and maybe a button that opens a text file (`fileButton` , `accept`). The Splat equation toy of Level 3 is one; so is the Word vectors toy, where you type *king* – *man* + *woman*.
- **Drawing:** `pad: { cols, rows, max, button }` adds a small grid you paint with a soft brush, and a button that hands the drawing to `read` as the text `"pad:..."` (one number for each cell, row by row). The CNN toy's "3D, draw a digit" view reads the digit you draw.
- **Opening a picture:** `media: { accept }` adds an **Open a file...** button, a field for a web address, the page buttons (or play and pause for a video) and a way back to the sample. Level 5's picture toys use it.



The input panel of a picture toy, in the Toy tab. Files are read on your device; nothing is uploaded.

For typing and drawing, `read(text, fileName)` turns what came in into option values, or throws an `Error` whose message the panel shows in place; and `shown()` says what is showing now. Keep what goes into options small: options are saved in the toy's link. Here is a whole toy with a typed input, a row of bars for the numbers you give it:

```
// A small toy with its own typed input, for Level 5 of the Tinkerer's Manual:
// type some numbers and the toy stands them up as bars.
const LIMIT = 12;
const COLORS = ["#e8384f", "#ffc233", "#1fb59a", "#4dabf7"];

export const RECIPES = {
```

```

"little-bars": {
  options: [
    {
      key: "numbers",
      label: "Your numbers",
      type: "text",
      default: "3 1 4 1 5 9 2 6",
      hidden: true,
    },
  ],
  input: {
    title: "Your own numbers",
    placeholder: "3 1 4 1 5 9 2 6",
    button: "Show them",
    fileButton: "Open a text file...",
    accept: ".txt,.csv",
    note: `Type up to ${LIMIT} numbers from 0 to 20, separated by spaces or commas, or open a text
file with them.` ,
    // Turns the typed text into option values, or throws a message that the panel shows.
    async read(text) {
      const list = String(text).match(/-?\d+(\.\d+)?/g) || [];
      if (list.length === 0) throw new Error("Type some numbers, like 3 1 4 1 5.");
      if (list.length > LIMIT) throw new Error(`That is more than ${LIMIT} numbers.`);
      if (list.some((n) => n < 0 || n > 20)) throw new Error("Use numbers from 0 to 20.");
      return { numbers: list.join(" ") };
    },
    shown: () => "Your numbers",
  },
  build(k, o) {
    const values = String(o.numbers).split(" ").map(Number);
    values.forEach((v, i) => {
      const h = 0.05 + v / 10;
      k.add(k.box(0.14, h, 0.14), {
        pos: [(i - (values.length - 1) / 2) * 0.2, h / 2, 0],
        color: COLORS[i % COLORS.length],
      });
    });
  },
},
},
};

```

The Screen and the Gaussian splatting toy

Two more toys put these to work: the Screen on the Pictures and pages shelf, and Gaussian splatting on the AI and computing shelf.

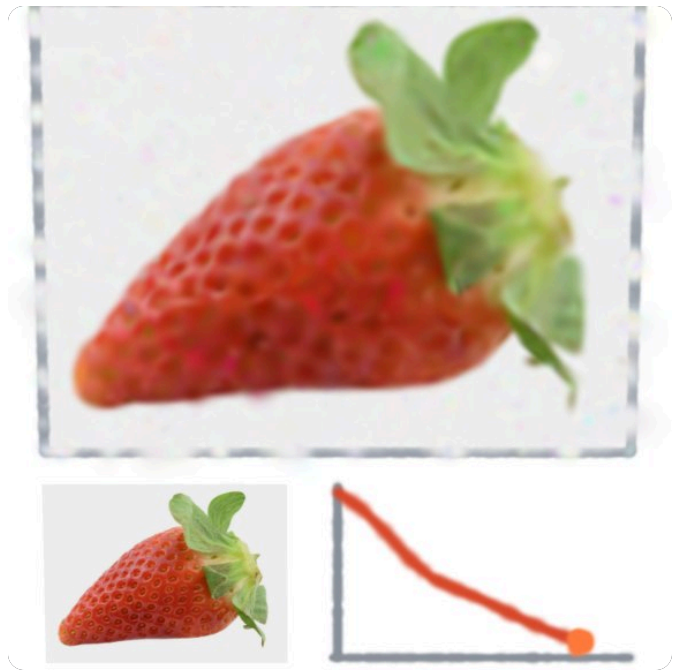
The **Screen** shows a video or a GIF on a screen in one of four styles: an old TV, a flat TV, a cinema with curtains and a hologram. Its picture is one sheet with `method: "screen"`; the knobs, the curtains and the glass are ordinary kit parts. A tap switches it on, and then plays and pauses; the speaker button in the top bar turns the sound on. You can open your own video or GIF.

The **Gaussian splatting** toy lets you watch how a photo becomes splats. Its **Training** view fits 2,400 flat (two-dimensional) Gaussians to a photo (a strawberry,

or your own), by gradient descent in a background worker: the splats start as random specks and, step by step, they slide, stretch and change color until together they are the picture. It draws them exactly as the site's renderer does (front to back, with the bell of Level 1), so what you see is the real fit. **One splat** lets you set the sizes, the opacity and the color of a single splat and turn it (Level 1's numbers); **Many splats** builds a small rubber duck from kit shapes and then shrinks every splat to a dot, so you see the splats that make it; and **Sorting** shows the splats of a ball appearing one by one, in the back-to-front order a camera draws them.

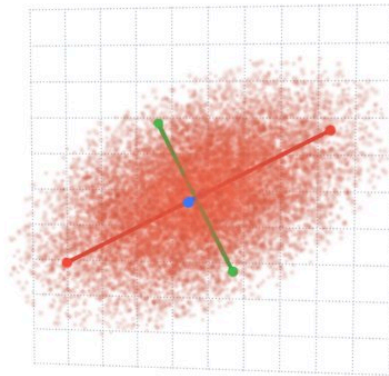


The Screen (old TV)



Gaussian splatting, training

Video frame from "Big Buck Bunny" © Blender Foundation, CC BY 3.0. Strawberry photo by Joselodos, CC0.



The One splat view: a single splat with its three sizes drawn as axes (Level 1's s_1 , s_2 and s_3).

Steering the view: the top bar and the tilt lock

Three round buttons sit in the top bar, beside the sound and help buttons. They control how you look at the toy on show, and none of them changes the toy.



- **Reset view** (the circular arrow) puts the camera back where the toy starts. The **R** key and a double tap do the same, and so does the button in the Tools tab.
- **Tilt lock** (the padlock) makes a drag turn a toy only left and right, so a flat toy such as a page, a photo or a screen stays facing you. Zooming, panning and pinching still work. Your choice is remembered for that toy for the rest of your visit. In a recipe, it is the flag `tiltLock: true`, and every toy on the Pictures and pages shelf sets it. A link keeps the camera it was saved with, and the lock then holds that tilt.
- **Turntable** (the record player) switches the slow turn that a toy does when nobody is touching it. It is the same setting as **Turntable when idle** in the Toy tab, and this device remembers it. A recipe can opt out with `turntable: false`, as a page does. If your device asks for less motion, the turn stays off and the button is disabled.

Two fingers on a touch screen zoom. They roll the toy only after a clear twist (about 17 degrees) and turn it only when they move together, so a normal pinch never spins what you are looking at.

What comes later: Code a toy

The splat equation language can place and color splats, but it can't give a toy parts, controls or a tap of its own. For that you need a full recipe (Level 2). Splashery's plan includes a **Toy Workshop** where you will be able to write one: a **Code a toy** editor that builds and shows your recipe as you type.

Sharing those toys will work differently from sharing equations. A recipe is real JavaScript, and a page that ran a stranger's JavaScript straight from a link could be made to do things its visitor never agreed to. So a coded toy will be shared as a **submission**: it is checked and read by a person before it joins the shelf, the way every toy in Splashery is today. Equations stay the quick way to share; recipes are the deep one.

Until then, everything in Level 2 is in the source code, which is open under the MIT license: github.com/ryanjosephkamp/splashery.

Quick reference

One splat

$$G(p) = \exp(-\frac{1}{2} (p - \mu)^T \Sigma^{-1} (p - \mu)), \quad \Sigma = R S S^T R^T$$

$$\Sigma' = J W \Sigma W^T J^T, \quad C \leftarrow a c + (1 - a) C \text{ (back to front)}$$

A splat program

```
x = ...      y = ...      z = ...      # in u, v and t (y is up)
u = 0 .. 2pi # ranges: numbers only
v = 0 .. pi  # default 0 .. 1
hue = ...    # 0 red, 1/3 green, 2/3 blue
r = ...      g = ...      b = ...      # or colors from 0 to 1
size = 0.05  # about the bell's  $\sigma$ 
count = 4000 # 100 to 10,000
spread = grid # or random
```

Functions: sin, cos, tan, asin, acos, atan, sinh, cosh, tanh, exp, log (ln), log10, sqrt, abs, floor, min, max. **Constants:** pi, e. **Time:** t runs from 0 to 2π per tap. At most 120 characters a field.

A picture toy

```
pictures: { sample: () => "assets/...", accept: ["pdf", "image", "gif", "video"] },
input: { title, media: { accept } },      // or placeholder, button, read, shown; or pad
tiltLock: true,                          // a drag turns it only left and right
build(k) { k.sheet({ id, center, width, height, leaf }); k.spine({ at, axis, dir }); },
drive(t, c, out, info) {
  out.sheets[id] = { page, visible };      // pages count from 0
  out.leaves[slot] = { angle, curl };      // radians; 0 flat, pi turned over
  info.data.pictures                        // page, count, go(n), next(), prev(), togglePlay()
}
```

The Tinkerer's Manual, first published September 28, 2026 and updated September 29, 2026. Part of [Splashery](#), a toy box of 3D Gaussian splats that runs in your browser. Every picture here was made with Splashery itself.